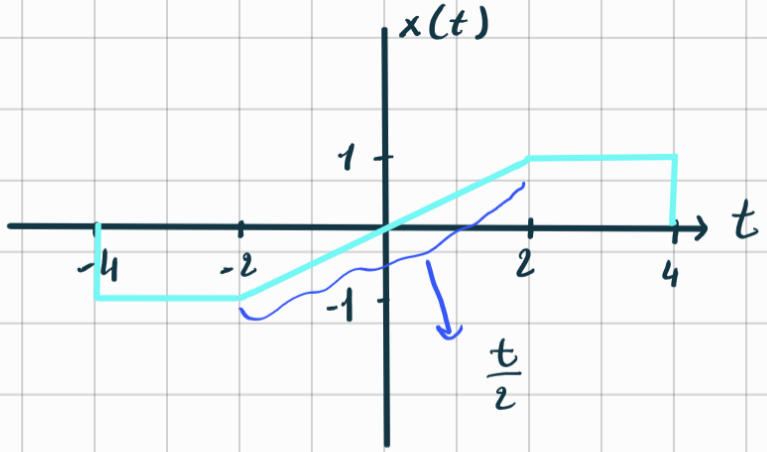
1)



a) x(t)'nin fourier transformunu bularak genlik ve faz spektrumunu çizdiriniz.

Periyodik olmayan (aperiyodik) işaretler için fourier dönüşümü aşağıdaki gibi tanımlıdır.

$$X(f) = FF\{x(t)\} = \int_{-\infty}^{\infty} x(t)\, e^{-j2\pi ft}\, dt$$

$$x(t) = \begin{cases} -1, & -4 < t < -2 \\ \frac{t}{2}, & -2 \le t < 2 \\ 1, & 2 \le t < 4 \\ 0, & \text{diger} \end{cases}$$

$$X(f) = \int_{-\infty}^{\infty} x(t)\, e^{-j2\pi ft}\, dt$$

$$= -\int_{-4}^{-2} e^{-j2\pi ft}\, dt + \frac{1}{2}\int_{-2}^{2} t\, e^{-j2\pi ft}\, dt + \int_{2}^{4} e^{-j2\pi ft}\, dt$$

$$= -\left(\frac{e^{-j2\pi ft}}{-j2\pi f} \right)\Bigg|_{-4}^{-2} + \frac{1}{2}\int_{-2}^{2} t\, e^{-j2\pi ft}\, dt - \left(\frac{e^{-j2\pi ft}}{j2\pi f} \right)\Bigg|_{2}^{4}$$

$$= \frac{1}{j2\pi f}\left(e^{-j2\pi f2} - e^{-j2\pi f4} \right) - \frac{1}{j2\pi f}\left(e^{-j2\pi f4} - e^{-j2\pi f2} \right) + \frac{1}{2}\int_{-2}^{2} t\, e^{-j2\pi ft}\, dt$$

$$\cos wt = \frac{1}{2}\left(e^{jwt} + e^{-jwt} \right)$$

$$\sin wt = \frac{1}{2j}\left(e^{jwt} - e^{-jwt} \right)$$

$$= \frac{1}{j2\pi f}\left(e^{j4\pi f} - e^{j8\pi f} - e^{-j8\pi f} + e^{-j4\pi f}\right) + \frac{1}{2}\int_{-2}^{2} t e^{-j2\pi ft}dt$$

$$= \underbrace{\frac{1}{j\pi f}\left(\cos(4\pi f) - \cos(8\pi f)\right)}_{I_1} + \frac{1}{2}\underbrace{\int_{-2}^{2} t e^{-j2\pi ft}dt}_{I_2}$$

$$I_2 = \int_{-2}^{2} t e^{-j2\pi ft}\,dt \qquad \text{Laplace}$$

$t = u$, $dt = du$; $v\,dv = e^{-j2\pi ft}\,dt$, $v = \frac{e^{-j2\pi ft}}{-j2\pi f}$

$$= uv - \int v\,du = \frac{t\cdot e^{-j2\pi ft}}{-j2\pi f}\Big|_{-2}^{2} + \int_{-2}^{2}\frac{e^{-j2\pi ft}}{+j2\pi f}\,dt$$

$$= \frac{1}{-j2\pi f}\left(2e^{-j4\pi f} + 2e^{j4\pi f}\right) + \frac{1}{j2\pi f}\cdot\frac{-1}{j2\pi f}\left(e^{-j2\pi ft}\right)\Big|_{-2}^{2}$$

$$= \frac{-1}{j\pi f}\left(\cos(4\pi f)\right) + \frac{1}{1\cdot 4\pi^2 f}\left(e^{-j4\pi f} - e^{+j4\pi f}\right)$$

$\sin\omega t = \frac{1}{2j}\left(e^{j\omega t} - e^{-j\omega t}\right)$

$$= \frac{-1}{j\pi f}\left(\cos(4\pi f)\right) + \frac{j}{4\pi^2 f j}\left(e^{-j4\pi f} - e^{j4\pi f}\right)$$

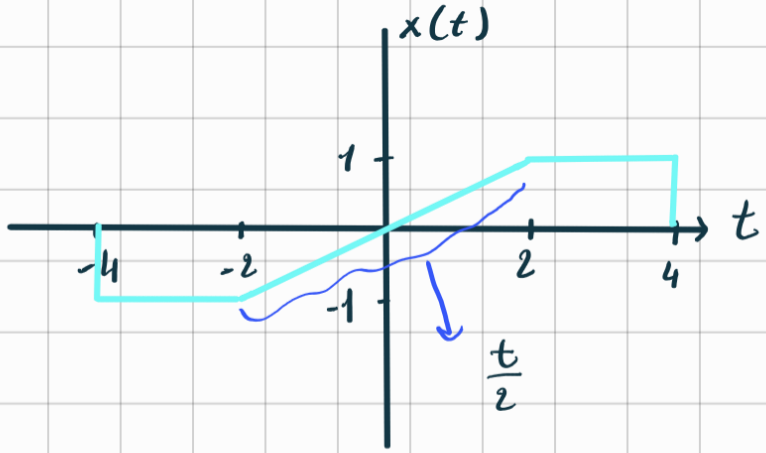
$$= -\frac{1}{j\pi f}\left(\cos(4\pi f)\right) + \frac{j}{2\pi^2 f}\sin(4\pi f)$$

$$I = I_1 + \frac{1}{2}I_2$$

$$= \frac{1}{j\pi f}\left(\cos(4\pi f) - \cos(8\pi f)\right) - \frac{1}{j2\pi f}\cos(4\pi f) + \frac{1}{8\pi^2 f}\cos(4\pi f)$$

$$= \frac{1}{j2\pi f}\cos(4\pi f) - \frac{1}{j\pi f}\cos(8\pi f) + \frac{1}{8\pi^2 f}\cos(4\pi f)$$

$$= \frac{-j}{2\pi f}\cos(4\pi f) + \frac{1}{8\pi^2 f}\cos(4\pi f) + \frac{j}{\pi f}\cos(8\pi f)$$



a) x(t)'nin fourier transformunu bularak genlik ve faz spektrumunu çizdiriniz.

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$$\frac{e^{j4\pi f} - e^{j8\pi f}}{2\pi fj}$$

$$I_2 = \frac{1}{2}\int_{-2}^{2} t e^{-j2\pi f t}\, dt$$

Laplace

$t = u$ $\quad dt = du$

$dv = e^{-j2\pi f t}\, dt$ $\quad v = \frac{e^{-j2\pi f t}}{-j2\pi f}$

$t = u$ $\quad dv = e^{-j2\pi f t}\, dt$

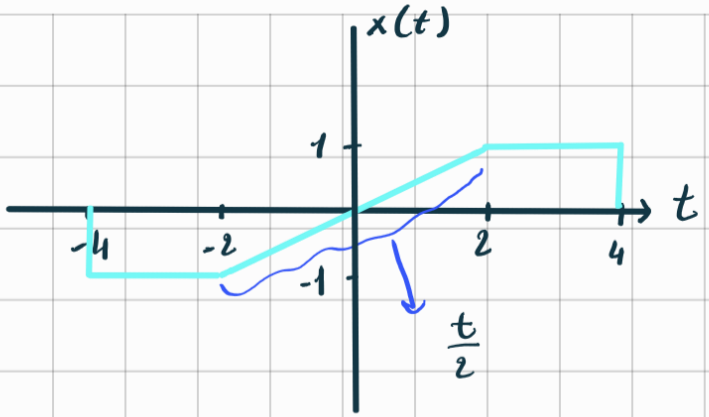
$\Rightarrow dt = du$ $\quad v =$

$$\cos wt = \frac{1}{2}\left(e^{jwt} + e^{-jwt}\right)$$
$$\sin wt = \frac{1}{2j}\left(e^{jwt} - e^{-jwt}\right)$$

$$-\frac{e^{-j2\pi f}}{2\pi f j} - \frac{e^{-j4\pi f}}{8\pi^2 f^2} - \frac{e^{4\pi f j}}{2f\pi j} + \frac{e^{4\pi f j}}{8\pi^2 f^2}$$

$$-\frac{\cos(4\pi f)}{\pi f j} +$$

Doğru



a) x(t)'nin fourier transformunu bularak genlik ve faz spektrumunu çizdiriniz.

Periyodik olmayan (aperiyodik) işaretler için fourier dönüşümü aşağıdaki gibi tanımlıdır.

$$X(f) = FF\{x(t)\} = \int_{-\infty}^{\infty} x(t)\, e^{-j2\pi ft}\, dt$$

$$x(t) = \begin{cases} -1, & -4 < t < -2 \\ \frac{t}{2}, & -2 \le t < 2 \\ 1, & 2 \le t < 4 \\ 0, & \text{diger} \end{cases}$$

$$X(f) = \int_{-\infty}^{\infty} x(t)\, e^{-j2\pi ft}\, dt$$

$$X(f) = \underbrace{\int_{-4}^{-2} -e^{-j2\pi ft}\, dt}_{I_1} + \underbrace{\int_{-2}^{2} \frac{t}{2} e^{-j2\pi ft}\, dt}_{I_2} + \underbrace{\int_{2}^{4} e^{-j2\pi ft}\, dt}_{I_3}$$

$$I_1 = \int_{-4}^{-2} -e^{-j2\pi ft}\, dt = \left(\frac{+e^{-j2\pi ft}}{+j2\pi f} \right) \Bigg|_{-4}^{-2}$$

$$= \frac{1}{j2\pi f} \left(e^{+j4\pi f} - e^{j8\pi f} \right)$$

$$I_3 = \int_2^4 e^{-j2\pi ft}\,dt = \left(\frac{e^{-j2\pi ft}}{-j2\pi f}\right)\Bigg|_2^4 = \frac{1}{-j2\pi f}\left(e^{-j8\pi f} - e^{-j4\pi f}\right)$$

$$= \frac{1}{j2\pi f}\left(e^{-j4\pi f} - e^{-j8\pi f}\right)$$

$$I_2 = \int_{-2}^{2} \frac{t}{2} e^{-j2\pi ft}\,dt = \frac{1}{2}\int_{-2}^{2} t\, e^{-j2\pi ft}\,dt$$

kısmi integrasyon $\int u\,dv = uv - \int v\,du$

$t = u$ $\qquad dv = e^{-j2\pi ft}dt$

$dt = du$ $\qquad v = -\dfrac{e^{-j2\pi ft}}{j2\pi f}$

$$I_2 = \frac{1}{2}\left[\left(t\,\frac{-e^{-j2\pi ft}}{j2\pi f}\right)\Bigg|_{-2}^{2} + \frac{1}{j2\pi f}\int_{-2}^{2} e^{-j2\pi ft}\,dt\right]$$

$$= \frac{1}{2}\left[\left(\frac{-2e^{-j4\pi f}}{j2\pi f} - \frac{2e^{j4\pi f}}{j2\pi f}\right) + \frac{1}{j2\pi f}\left(\frac{e^{-j2\pi ft}}{-j2\pi f}\right)\Bigg|_{-2}^{2}\right]$$

$$= \frac{1}{j2\pi f}\left[-e^{-j4\pi f} - e^{j4\pi f} - \frac{1}{j4\pi f}\left(e^{-j4\pi f} - e^{j4\pi f}\right)\right]$$

$$= \frac{1}{j2\pi f}\left[-e^{-j4\pi f} - e^{j4\pi f} + \frac{1}{j4\pi f}\left(e^{j4\pi f} - e^{-j4\pi f}\right)\right]$$

$\cos wt = \frac{1}{2}\left(e^{jwt} + e^{-jwt}\right)$

$\sin wt = \frac{1}{2j}\left(e^{jwt} - e^{-jwt}\right)$

$$X(f) = I_1 + I_3 + I_2$$

$$= \frac{1}{j2\pi f}\left(e^{j4\pi f} - e^{j8\pi f} + e^{-j4\pi f} - e^{-j8\pi f} - e^{-j4\pi f} - e^{j4\pi f} + \frac{1}{j4\pi f}\left(e^{j4\pi f} - e^{-j4\pi f}\right)\right]$$

$$= \frac{1}{j2\pi f}\left[\left(e^{j8\pi f} + e^{-j8\pi f}\right) + \frac{1}{j4\pi f}\left(e^{j4\pi f} - e^{-j4\pi f}\right)\right]$$

$$= \frac{1}{j2\pi f}\left[-2\cos(8\pi f) + \frac{1}{2\pi f}\sin(4\pi f)\right]$$

ve $\text{sinc}(t) = \begin{cases} 1 & x=0 \\ \frac{\sin(\pi t)}{\pi t} & \text{diğer} \end{cases}$

$$= \frac{1}{j2\pi t}\left[-2\cos(8\pi t) + \frac{2}{4\pi t}\sin(4\pi t)\right]$$

$$= \frac{1}{j2\pi t}\left[-2\cos(8\pi t) + 2\,\text{sinc}(4t)\right]$$

$$= \frac{j}{2\pi t}\left[2\cos(8\pi t) - 2\,\text{sinc}(4t)\right] = \frac{j}{\pi t}\left[\cos(8\pi t) - \text{sinc}(4t)\right]$$

b) $y(t) = x(t-5)$ işaretinin Fourier transformunu bularak genlik vefaz spektrumunu çizdiriniz.
Fourier transformun zamanda öteleme özelliği aşağıdaki gibidir.

$x(t) \xleftrightarrow{FF} X(f)$ olmak üzere
$x(t-t_0) \xleftrightarrow{FF} e^{-j2\pi f t_0} X(f)$

$$X(f) = \frac{j}{\pi t}\left[\cos(8\pi t) - \text{sinc}(4t)\right]$$

ve $y(t) = x(t-5)$

$$\Rightarrow Y(f) = e^{-j2\pi f 5} X(f) = e^{-j10\pi f} X(f)$$

$$= \frac{e^{-j10\pi f}}{\pi t}\left[\cos(8\pi t) - \text{sinc}(4t)\right]$$

c) $y(t) = x(t)\, e^{j10\pi t}$ işaretinin Fourier transformunu bularak genlik ve faz spektrumlarını çiziniz.

Frekansta öteleme: $x(t)\, e^{j2\pi f_0 t} \xleftrightarrow{FF} X(f-f_0)$

ve $g(t) = x(t)\, e^{j2\pi \cdot 5 \cdot t} \Rightarrow f_0 = 5$

$\Rightarrow Y(f) = X(f-5)$

$$Y(f) = \frac{j}{2\pi(f-5)} \left[\cos(8\pi(f-5)) - \text{sinc}(4(f-5))\right]$$

$$Y(f) = \frac{j}{2\pi f - 10\pi} \left[\cos(8\pi f - 40\pi) - \text{sinc}(4f-20)\right]$$

d) $y(t) = \frac{1}{3} x(t/3)$ işaretinin fourier transformunu bularak genlik ve faz spektrumunu çizdiriniz.

Zamanda ölçeklendirme özelliği: $x(at) \xleftrightarrow[a\neq 0]{FF} \frac{1}{|a|} X(f/a)$

$a>1 \Rightarrow$ işaret frekansta genişler

$0<a<1$ işaret frekansta daralır

Lineerlik özelliği: $\alpha x_1(t) + \beta x_2(t) \xleftrightarrow{FF} \alpha X_1(f) + \beta X_2(f)$

Bu özellikleri kullanarak soruyu çözelim:

$$x\left(\frac{1}{3}t\right) \xleftrightarrow{FF} \frac{1}{\left|\frac{1}{3}\right|} X\left(\frac{f}{\frac{1}{3}}\right) = 3X(3f)$$

$$Y(f) = \frac{1}{3} 3X(3f) = X(3f)$$

$$Y(f) = \frac{j}{2\pi(3f)}\left[2\cos(8\pi(3f)) - 4\,\mathrm{sinc}(4(3f))\right]$$

$$Y(f) = \frac{j}{6\pi f}\left[2\cos(24\pi f) - 4\,\mathrm{sinc}(12 f)\right]$$

e) Çıkış işareti $y(t) = \mathrm{sgn}(t)$ olarak verilen LTI sistemin frekans ve dürtü cevabını bulunuz. Bulduğunuz frekans cevabının genlik ve faz spektrumunu çizdiriniz

$$y(t) = \mathrm{sgn}(t) = \begin{cases} 1, & t>0 \\ 0, & t=0 \\ -1, & t<0 \end{cases} = u(t) - u(-t)$$

$$u(t) = \int_{-\infty}^{t} \delta(z)\,dz$$

$$\mathcal{F}\{\delta(t)\} = \int_{-\infty}^{\infty} \delta(t) e^{-j2\pi f t}\,dt = \int_{-\infty}^{\infty} \delta(t) e^{-j2\pi f\cdot 0}\,dt = \int_{-\infty}^{\infty} \delta(t)\,dt = 1$$

$$x(t) = \delta(t) \xleftrightarrow{\mathcal{F}} X(f) = 1$$

Integral alma özelliği: $\int_{-\infty}^{t} x(z)\,dz \xleftrightarrow{\mathcal{F}} \frac{X(f)}{j2\pi f} + \frac{1}{2}X(0)\,\delta(f)$

$$\mathcal{F}\{u(t)\} = \frac{1}{j2\pi f} + \frac{1}{2}\delta(f)$$

ölçeklendirme özelliği: $x(at) \longleftrightarrow \frac{1}{|a|}x(f/a)$

$$\mathcal{F}\{\mathrm{sgn}(t)\} = \frac{1}{j2\pi f} + \frac{1}{2}\delta(f) - \frac{1}{|-1|}\left[\frac{1}{j2\pi(-f)} + \frac{1}{2}\delta(-f)\right]$$

$$= \frac{1}{j2\pi f} + \frac{1}{2}\delta(f) + \frac{1}{j2\pi f} - \frac{1}{2}\delta(f)$$

$\delta(f) = \delta(-f)$ çift fonksiyon

$$Y(f) = \frac{1}{j\pi f} \;//$$

$$x(t) * h(t) = g(t)$$
$$X(f) \cdot H(f) = Y(f)$$

$$\left(\frac{j}{\pi f} \left[\cos(8\pi f) - \operatorname{sinc}(4f) \right] \right) \cdot H(f) = \frac{1}{j\pi f}$$

$$H(f) = \frac{\frac{1}{j\pi f}}{-\frac{1}{j\pi f}\left[\cos(8\pi f) - \operatorname{sinc}(4f)\right]}$$

$$= \frac{1}{\frac{\sin 4\pi f}{4\pi f} - \cos(8\pi f)}$$

$\cos\omega t = \frac{1}{2}\left(e^{j\omega t} + e^{-j\omega t}\right)$
$\sin\omega t = \frac{1}{2j}\left(e^{j\omega t} - e^{-j\omega t}\right)$

$4\pi f = a$ olsun

$$= \frac{1}{\frac{\sin a}{a} - \cos 2a} = \frac{1}{\frac{\sin a - a\cos a}{a}}$$

$$= \frac{a}{\sin a - a\cos 2a} = \frac{a}{\frac{1}{2j}\left(e^{ja} - e^{-ja}\right) - \frac{a}{2}\left(e^{j2a} + e^{-j2a}\right)}$$

=!!

2) $x[n] = (0,8)^n u[n]$

a) $X(e^{jw}) = \sum_{k=-\infty}^{\infty} x[n]\, e^{-jwn} = \sum_{n=-\infty}^{\infty} (0,8)^n u[n]\, e^{-jwn}$

$= \sum_{k=0}^{\infty} (0,8)^n e^{-jwn} = \sum_{n=0}^{\infty} (0,8\, e^{-jw})^n = \frac{1}{1 - 0,8 e^{-jw}}$

b) $y[n] = x[n-5]$ $\quad Y(e^{jw}) = ??$

Zamanda ötelelem özelliği: $x[n-n_0] = e^{-jwn_0} X(e^{jw})$

$\Rightarrow Y(e^{jw}) = e^{-jw5} X(e^{jw})$

$= \frac{e^{-j5w}}{1 - 0,8\, e^{-jw}}$

c) $y[n] = n x[n] \Rightarrow Y(e^{jw}) = ??$

Frekansta türev özelliği: $n x[n] \xleftrightarrow{FT} j \frac{dX(e^{jw})}{dw}$

$Y(e^{jw}) = j \frac{d X(e^{jw})}{dw}$

Nasıl türev alınır ?? w'ye göre mi, e^{jw}'ye göre mi ?? Bilemem.

$\frac{dX(e^{jw})}{dw} = \frac{d}{dw} \frac{1}{1 - 0,8 e^{-jw}}$ zincir kuralını kullanalım $= \frac{d}{du} \frac{1}{u} \frac{du}{dw}$

$u = 1 - 0,8\, e^{-jw}$ olsun

$\frac{d}{du}\left(\frac{1}{u}\right) = -\frac{1}{u^2}$

$\Rightarrow \frac{dX(e^{jw})}{dw} = \frac{-\frac{d}{dw}(1 - 0,8\, e^{-jw})}{(1 - 0,8\, e^{-jw})^2}$

$= -\frac{1}{(1 - 0,8\, e^{-jw})^2} \cdot (0 - (-0,8j)\, e^{-jw})$

$= -\frac{(0,8j)\, e^{-jw}}{(1 - 0,8\, e^{-jw})^2}$

d) $y[n] = x[-n] \Rightarrow Y(e^{j\omega}) = ?$

Zamanda katlama özelliği: $x[-n] = X(e^{-j\omega})$

$$Y(e^{j\omega}) = X(e^{-j\omega}) = \frac{1}{1-0{,}8e^{j\omega}}$$

e) $y[n] = x[n]\cos(0{,}1\pi n) \Rightarrow Y(e^{j\omega}) = ?$

$$\cos(0{,}1\pi n) = \frac{e^{0{,}1\pi nj} + e^{-0{,}1\pi nj}}{2}$$

ve Frekansta öteleme özelliği: $e^{j\omega_0 n}x[n] \overset{FF}{\longleftrightarrow} X(e^{j(\omega-\omega_0)})$

$$y[n] = \frac{1}{2}\left(x[n]e^{0{,}1\pi nj} + x[n]e^{-0{,}1\pi nj}\right)$$

$$Y(f) = \frac{1}{2}\left[X(e^{j(\omega-0{,}1\pi)}) + X(e^{j(\omega+0{,}1\pi)})\right]$$

$$= \frac{1}{2}\left[\frac{1}{1-0{,}8e^{j(\omega-0{,}1\pi)}} + \frac{1}{1-0{,}8e^{j(\omega+0{,}1\pi)}}\right]$$

f. Çıkış işareti $y[n] = \delta[n-5]$ olarak verilen LTI sistemin frekans ve dürtü cevabını bulunuz. Bulduğunuz frekans cevabının genlik ve faz spektrumunu çiziniz.

$\delta[n] \overset{FF}{\longleftrightarrow} 1$

ve Zamanda öteleme özelliği: $x[n-n_0] \overset{FF}{\longleftrightarrow} e^{-j\omega n_0}X(e^{j\omega})$

$\Rightarrow \delta[n-5] = e^{-j\omega 5} = e^{-j5\omega}$

$Y(e^{j\omega}) = e^{-j5\omega}$

$x[n] * h[n] = y[n]$

$X(e^{j\omega}) \cdot H(e^{j\omega}) = Y(e^{j\omega})$

$$\frac{1}{1-0{,}8e^{-j\omega}} \cdot H(e^{j\omega}) = e^{-j5\omega}$$

$$H(e^{j\omega}) = e^{-j5\omega} - 0{,}8\, e^{-j\omega} \cdot e^{-j5\omega}$$

$$= e^{-j5\omega} - 0{,}8\, e^{-j6\omega}$$

$$\delta[n-n_0] \longleftrightarrow e^{-j\omega n_0}$$

$$\Rightarrow h[n] = \delta[n-5] - 0{,}8\,\delta[n-6]$$